

SKF Deep Groove Ball Bearings - Detailed Guide

SKF Deep Groove Ball Bearings - Detailed Overview

Deep groove ball bearings are the most widely used rolling bearings. They are designed to carry radial loads and moderate axial loads in both directions.

Construction: These bearings consist of an inner ring, outer ring, steel balls, and a cage that keeps the balls evenly spaced.

Applications: - Electric motors - Pumps - Gearboxes - Household appliances - Automotive components

Advantages: - Low friction - High speed capability - Low maintenance - Long service life

Operating Principle: The deep groove geometry allows smooth rolling motion with minimal resistance. The load is distributed evenly across the balls.

Variants: - Single row deep groove ball bearings - Sealed and shielded versions - High-speed variants

Maintenance: Proper lubrication is essential. Grease or oil lubrication is used depending on operating conditions.

Failure Causes: - Improper lubrication - Contamination - Misalignment